About the UWBe Impact Atlas

One Billion Drops · Climate Action · United Way Bengaluru
Prepared by WE360 – Water and Environment Consultants · Version 2 · 15 October 2026

The atlas is an interactive map of the One Billion Drops (OBD) programme. It shows where percolation wells were built, who funded them, whether the monitored wells hold water month by month, and how that compares with rainfall and with Bengaluru's groundwater situation. It is meant for relationship managers, donors and partners who want to see the programme's footprint and evidence in one place.

Web address: `https://kkb517.github.io/OBD_Impact_Assessment_Atlas_Oct_2026/`

1. The screen at a glance

The left panel holds the controls; the map fills the rest of the screen. On a phone the map sits on top and the panel scrolls below it. From top to bottom, the panel contains:

Section	What it does
About	Opens this document.
Filter by donor and financial year	Three drop-down filters (Donor, Financial year, BBMP zone) and a summary card that updates as you filter. See section 2.
Monitoring month	Slider and play button to step through the monthly well surveys, May 2025 to September 2026. The map recolours the monitored wells for the chosen month. Three figures sit below the slider: share of monitored wells holding water, average depth of water standing in the wells, and that month's rainfall against its normal.
Rainfall and wells holding water	Monthly rainfall (bars), the 1991–2020 normal for each month (black ticks), and the share of monitored wells holding water (green line). Click any month to show it on the map.
Go to	Zooms to all of Bengaluru, one of the three study valleys (KC, Hebbal, Vrishabhavathi) or Lalbagh.
Layers	Switches each map layer on or off. The number beside each layer is how many features it holds. See section 3.
Legend	Colour keys for the wells, borewells, OBD sites, stage of extraction, infiltration tests and survey.
Basemap	Grey (default; follows light or dark mode), Streets, Satellite, or None.
Context	Annual rainfall since 2010 against the long-term normal, and wells completed per year.

Click any point or area on the map for details. Hover over boundaries to see their names.

2. Filters and the summary card

The filters follow the way UWBe reports the programme: by donor and by financial year. They can be combined, for example "Morgan Stanley, FY 2024-25, Yelahanka zone". **Clear filters** in the summary card resets all three; **Zoom to selection** fits the map to the filtered sites and wells.

Percolation wells shown

All monitored wells (any month) shows every well in the month it was read: 1,263 wells in all, including wells read only once and the one-time Lalbagh survey (September 2025). This is the default, and its figures match the impact

presentation.

Wells monitored every month, from–to keeps only wells read in every month of the period you choose with the From and To menus. The default period is September 2025 to September 2026, the longest stretch in which the full set of about 740 wells was read: 501 wells were read in all 13 months, or 676 if you tick **Allow one missed month**. Earlier start dates leave fewer wells, because the regular set was built up between May and September 2025 (from May 2025: 53 wells every month, 247 with one miss). Use this option to compare like with like across months.

Monitoring started at different times in each valley: KC (Bommanahalli) in May 2025, Hebbal and Vrishabhavathi in August 2025. All three have been read together since September 2025. A period starting in May 2025 therefore covers KC wells only (53 read every month, 247 with one miss); one starting in August 2025 adds part of Hebbal and Vrishabhavathi.

Donor

Lists every donor that appears in the site list, the mapped well locations or the monthly monitoring (41 donors). Field records spell some names in several ways (for example "Morgon Stanley", "BROADCOM", "Swiss Ray Foundation"); these were standardised to the names in the site list before the filter was built.

Financial year

Indian financial year, April to March. It is taken from the "Year of Completion" in the UWBe site list: a site completed in March 2025 belongs to FY 2024-25, one completed in April 2024 also belongs to FY 2024-25. Years available: 2018-19, 2020-21, 2021-22, 2022-23, 2023-24 and 2024-25.

The site list on the map runs to March 2025. Monitored wells at parks that are not in that list are grouped under **"Not in site list (after Mar 2025)"**; most are FY 2025-26 sites. For these, the summary card counts only the monitored wells.

BBMP zone

The old eight BBMP zones, assigned by where each site or well lies on the zone map. Sites outside the BBMP boundary are grouped as "Outside BBMP". The new Greater Bengaluru Authority (GBA) corporations will be added as a layer once their boundary file is in the atlas folder.

What each filter applies to

Layer	Donor	Financial year	BBMP zone
OBD sites (to Mar 2025)	Yes	Yes	Yes
Individual well locations	Yes	Yes (from the site list)	Yes
Percolation wells (monitored monthly) and the month figures; also the "wells shown" option	Yes	Yes (from the site list)	Yes
Infiltration tests	Yes	No	No
Borewells, CGWB stations, survey, IISc wells, boundaries	Not filtered: they describe the aquifer and the area, not a donor's wells		

Choosing a donor or a year switches on the OBD sites layer so the selection is visible.

How the summary card is calculated

> **Estimated recharge capacity per well** = 4.27 m^3 per fill × 30 effective rainy days a year = **128.1 m^3 a year**. Effective rainy days are days with 10 mm of rain or more, from 30 years of IMD rainfall.
>
> **Tanker loads** = estimated recharge capacity ÷ 6,000 litres. **People supported** = estimated recharge capacity ÷ (75 litres × 365 days), the basic-needs figure used in the October 2026 impact presentation.

> Estimated recharge capacity is the volume the wells can send into the ground if each well fills and drains on every effective rainy day. It is an estimate based on the well design and rainfall, not a measured volume. Earlier documents called it "design recharge" or "design estimate"; these mean the same thing here. The 2026 infiltration tests support this figure: at the median measured rate, a full well drains in about 32 hours, so it can refill on most rainy days.

When a filter is active, the three figures under the month slider are recalculated for the filtered monitored wells only and a **Filtered** label appears. Without a filter, they match the presentation and exclude Lalbagh, which is monitored separately.

Change from the earlier atlas: version 1 expressed recharge as households at 135 litres per person per day. Version 2 uses 75 litres per person per day, to match the impact presentation. Both are valid; quote one consistently.

3. Layers

Layer	What it shows	Source and period
Percolation wells (monitored monthly) 739 regular wells	739 wells are read every month (11 or more of the 17 surveys). The layer also shows wells read only once or occasionally in the month they were read, including a one-time survey of 212 Lalbagh wells in September 2025. Each well is coloured by the depth of water standing in it in the chosen month: ● dry, then four blues from under 1 ft to 6 ft or more. Wells not read that month are hidden. Click for the well's full monthly record against rainfall.	UWBe monthly monitoring, May 2025 – Sep 2026
Borewells (UWBe) 52 wells	Depth to water below ground (purple ring). Darker blue is shallower. Click for the series; 16 wells also show the change since IISc measured them in 2015–17.	UWBe/WE360 fortnightly readings, Oct 2025 – Aug 2026
CGWB stations 68 stations	Government monitoring wells; blue ring = telemetric (several readings a day), dark ring = manual. Click for the monthly record since 2015.	CGWB via India-WRIS, 2015 – 2026
OBD sites (to Mar 2025) 419 sites, 8,081 wells	Parks and campuses with OBD wells. Size = number of wells; shade = financial year completed. Click for donor, wells, year and estimated recharge capacity.	UWBe site list (sites with coordinates)
Individual well locations 1,577 wells	GPS position of each well where it was recorded (a subset of all wells).	UWBe well register
Infiltration tests 27 tests	Double-ring infiltration tests. Colour = rate (under 1.5, 1.5–4, over 4 cm/h). Orange ring = March 2025 test. Click for rates, drain time and soil texture.	WE360 reports, Mar 2025 and Sep 2026
Citizen survey 2026 126 parks, 659 households	Survey responses grouped by park. Size = households; shade = share saying the wells helped groundwater. Click for awareness, flooding, soil moisture and upkeep.	UWBe citizen survey, Feb – Mar 2026
IISc wells 2007–13 615 wells	Historical depth-to-water survey across the city; click for average, shallowest and deepest reading.	IISc groundwater survey, 2007 – 2013
Stage of extraction 2025-26 6 taluks	Groundwater extracted as a share of annual recharge, by taluk. All six Bengaluru Urban taluks are over-exploited (106% to 395%). Click for resource, extraction and recharge from artificial structures.	CGWB/IIT Hyderabad INGRES, GEC 2025-26
Study sub-watersheds 6 units	The six study areas in the KC, Hebbal and Vrishabhavathi valleys. Click for the valley's recharge computation.	Watershed atlas boundaries (WE360 GIS)
Valleys, sub- and micro-watersheds	Hydrological boundaries. Click a valley for its recharge computation (study area and full implementation area).	Watershed atlas boundaries (WE360 GIS)
GBA corporations	Five corporations under the Greater Bengaluru Authority. Shown once the file is added.	OpenCity, Sep 2025
BBMP zones, wards, boundary; district	Administrative boundaries: the eight old zones, the 225 wards, the BBMP limit and Bengaluru Urban district.	BBMP; KGIS

4. Reading the numbers

- **Wells holding water follow rainfall.** Across the monitoring period, the share of wells holding water tracks rain in the preceding 30 days (correlation 0.74). Compare any month with its rainfall bar before drawing conclusions; September 2026, for example, was lower than September 2025 because July and August 2026 were dry.
- **"Normal"** is the average rainfall for that calendar month over 1991–2020 (about 840 mm a year), for the IMD grid cell that covers all six study areas. 2026 rainfall is IMD's provisional real-time data.
- **Borewells and CGWB stations** tap the deeper, semi-confined aquifer and respond to pumping nearby. Changes in single wells are not attributed to OBD.
- **Stage of extraction** is a taluk-wide figure. OBD wells are a small share of a taluk's water budget, so the map does not imply OBD changed any taluk's category.
- **Infiltration tests:** the March 2025 report values (0.2–0.6 cm/h) came from extended curve fits and textbook soil parameters; the raw field readings ended at 2.6–18 cm/h. Quote the September 2026 tests (median adopted rate 2.5 cm/h).
- **Survey results by park** come from small samples (often five households). Read the pattern across parks, not single values.
- **Well IDs:** field sheets write the same well in different ways ("PW-554" and "PW554", "D-77" and "D77"). These were merged when the ID matched after removing hyphens and spaces and the two positions were within 150 m. The raw data held 1,523 IDs; after merging there are 1,263, of which 739 are read every month.
- **Coverage:** the site list on the map covers wells completed to March 2025 (8,081 wells at 419 sites with coordinates). The programme total is 13,714 wells by March 2026.

> **Not shown yet:** water-quality results. Baseline tests of borewells (Feb 2026) and percolation wells (Jun–Jul 2026) are complete; results will be added as a pre- vs post-monsoon comparison after the October–November 2026 round.

5. Keeping the atlas up to date

The atlas is a static website on GitHub Pages: one `index.html` plus data files in the same folder. To update it, replace the relevant file in the GitHub repository (Add file → Upload files → Commit) and reload the page after about a minute. GitHub may serve the old copy for up to 10 minutes; adding `?v=2` to the address or pressing Ctrl+Shift+R shows the new one.

File	Contents
`percolation.json`	Monthly percolation-well readings with donor, financial year and zone
`borewells.json · cgwb.json · rainfall.json`	Borewell series, CGWB station series, monthly and annual rainfall
`parks.geojson · well_points.geojson`	OBD sites and individual well locations
`infiltration.geojson · survey.geojson · iisc0713.geojson · ingres.geojson`	Infiltration tests, survey by park, IISc 2007–13 wells, CGWB stage of extraction
`study_units, watersheds, subwatersheds, micro_watersheds, bbmp_*, wards, district (.geojson)`	Boundaries
`valley_summary.json · about.pdf · logo.png`	Valley recharge tables, this document, UWBe logo

`gba_corporations.geojson`	Optional; switches on the GBA layer when present

6. Sources

Central Ground Water Board. (2026). *Dynamic ground water resources of India, 2025-26* [INGRES]. Ministry of Jal Shakti, Government of India; IIT Hyderabad. https://ingres.iith.ac.in

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Hengl, T. (2018). *Soil texture classes (USDA system) for 6 soil depths at 250 m* (Version v02) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.1475451

OpenCity. (2025). *Greater Bengaluru Authority corporations delimitation 2025* [Data set]. https://data.opencity.in/dataset/greater-bengaluru-authority-corporations-delimitation-2025

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WE360 – Water and Environment Consultants. (2025). *Determination of soil infiltration capacity* [Technical reports to United Way Bengaluru, March 2025].

WE360 – Water and Environment Consultants. (2026). *Impact assessment of percolation wells: KC, Hebbal and Vrishabhavathi valleys* [Final reports to United Way Bengaluru, March 2026]; *Double-ring infiltration tests* [Reports, September 2026].

Basemaps: Esri World Light/Dark Gray Canvas, World Street Map and World Imagery (© Esri, HERE, Garmin, Maxar, © OpenStreetMap contributors). Map software: MapLibre GL JS 4.7.1.